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Machine Learning (MLOps) A Resilient System for Predictive Climate Risk Assessment

TeamNo-14

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Agenda

- Introduction
- Motivation (Why this title)
- Problem Definition
- Objectives
- Data set Description
- Risk Analysis
- Tools and Techniques used
- Phase I : Machine Learning Model and its analysis
- Phase II : Building MLOPs



Introduction

- Weather prediction plays a critical role in agriculture, disaster management, and planning.
- Machine learning models are widely used for predicting weather parameters.
- However, real-world weather data is dynamic and non-stationary.
- This project focuses on maintaining model reliability over time using MLOps.



Motivation (Why this title)

- **Cyclone & Heavy Rainfall Prediction(Real life scenario)**
 - Accurate weather prediction helps detect cyclones and heavy rainfall in advance.
 - Early warnings allow evacuation of people, suspension of fishing, and emergency preparedness.
 - This has significantly reduced loss of human life in coastal and flood-prone regions.
 - Weather patterns change due to environmental and external influences.
- Data collected today may not match historical data.
- ML models trained once may fail silently in production.
- MLOps helps in: Continuous monitoring, Detecting data drift,Ensuring long-term model reliability.



Problem Definition

Weather prediction models are trained using historical weather datasets. These datasets capture patterns and relationships that were true at the time of training. The model learns from this past data to make predictions about future weather conditions.

Over time, the characteristics of incoming weather data can change due to factors such as climate change, seasonal variations, or changes in data collection methods. As a result, the new data may no longer follow the same distribution as the data used during training.

This change in data distribution, known as data drift, can cause the model's prediction accuracy to gradually decrease. The model continues to rely on outdated patterns, which makes its forecasts less reliable.

If the model is not properly monitored, these incorrect or degraded predictions may go unnoticed for a long time. This can lead to poor decision-making and reduced trust in the weather prediction system.



Objectives

- To build a machine learning model for weather prediction.
- To design an MLOps pipeline that:
 - Monitors incoming data
 - Detects data drift over time
 - Alerts users when retraining is required
- To ensure transparency, traceability, and governance.



Dataset Description

- Historical weather dataset
- Contains meteorological features such as:
 - Temperature
 - Humidity
 - Pressure
 - Wind speed and direction
 - Cloud cover
- Target variable: weather-related prediction (e.g., rainfall)
- **Source of the dataset is** <https://www.kaggle.com/datasets/sumanthvrao/daily-climate-time-series-data>
<https://www.kaggle.com/datasets/pratikjadhav05/indian-weather-data>
- New data for predictions is gotten through Open-Meteo API



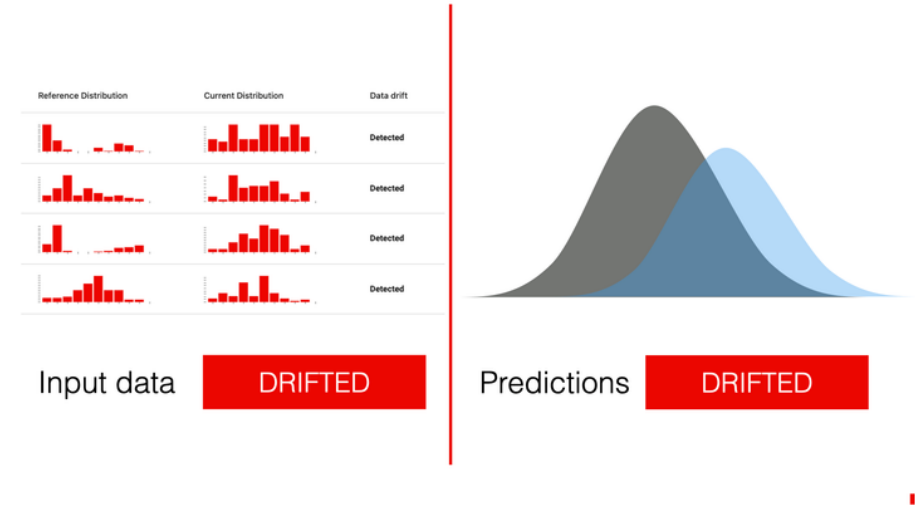
Risk analysis

Identified Risks

- Data distribution shift
- Model staleness
- Lack of monitoring
- Incorrect predictions in production

Impact

- Reduced prediction accuracy
- Loss of trust in ML system
- Poor decision-making





Phase I – ML Life Cycle



Exploratory Data Analysis

Data Cleaning and Preprocessing:

- Remove columns with dates and timestamps

Feature Engineering :

- remove all duplicate columns containing values in different units (ex Kph/Mph)
- Turn textual data regarding wind direction into numerical data through one hot encoding



List of algorithms chosen to build ML model

- Linear Regression
- XGboost
- Random forest



Metrics used to evaluate the ML model

Mean Absolute Error (MAE)

- MAE measures the average absolute difference between predicted and actual values.
- It shows how wrong the predictions are on average.
- Lower MAE means better model performance.
- Used to understand overall prediction accuracy without giving extra weight to large errors.

Root Mean Square Error (RMSE)

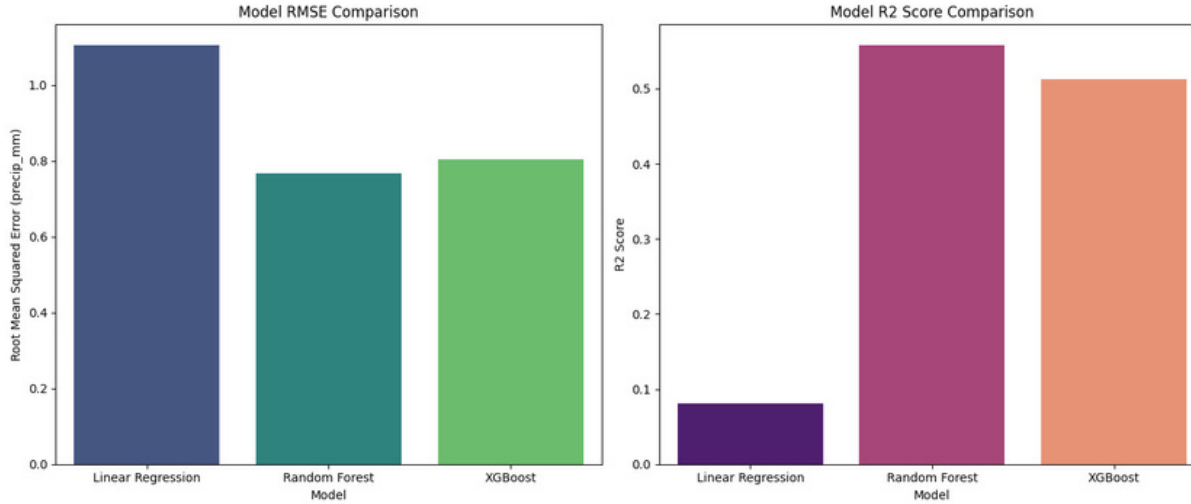
- RMSE measures the square root of the average squared errors.
- It penalizes large errors more than MAE.
- Useful when large prediction mistakes are critical, such as extreme weather values.
- Lower RMSE indicates a more reliable model.

R² Score (Coefficient of Determination)

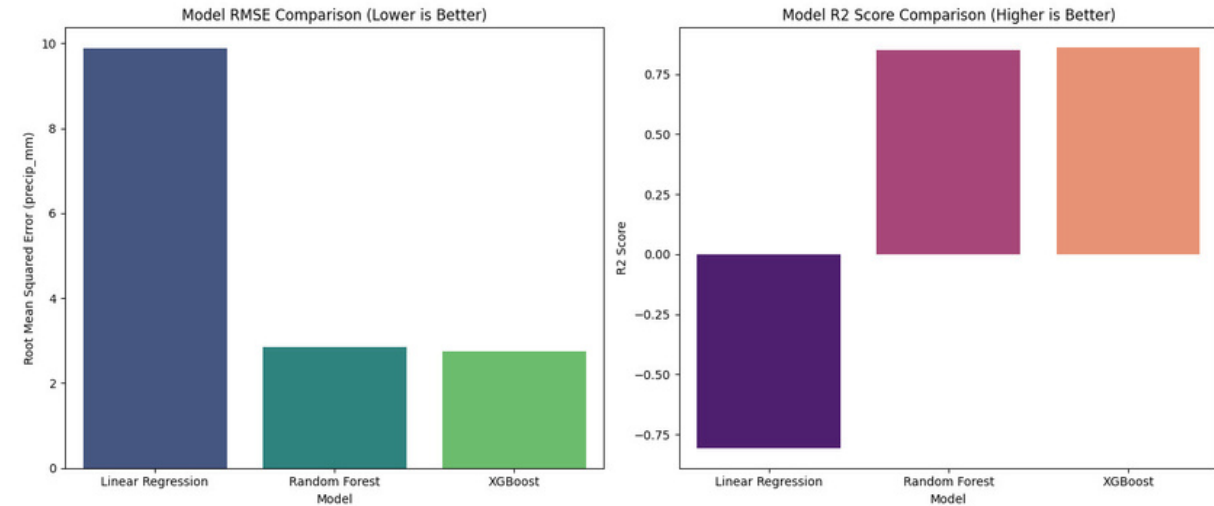
- R² shows how well the model explains the variability in the data.
- Value ranges from 0 to 1.
- Higher R² means the model fits the data better.
- Used to compare how well different models explain weather patterns.



Evaluation of ML model



**rain prediction model-
comparison of different
algorithms**



**temperature prediction model-
comparison of different
algorithms**



Identification of Best Model

based on matric evaluation

----Random forest for the temperature prediction model

----XGboost for the rain prediction model



Phase II – MLOPs Life Cycle



Phase II – MLOPs Life Cycle

	Component	MLOPs Tools
1	Data Management	DVC
2	Experiment Tracking and Version Control	ZenML
3	Automation and Pipeline Orchestration	ZenML
4	Model Deployment & Serving	Docker, FastAPI
5	Monitoring & Governance and collaboration	Evidently AI



Phase II – MLOPs Life Cycle

Monitoring and Governance

Goal:

Ensure model reliability when data changes over time.

- Incoming data continuously monitored
- Comparison with training data
- Detection of:
 - Feature drift
 - Prediction drift
- Alerts generated for human decision-making



Phase II – MLOPs Life Cycle

Detailed Explanation

1. Data Management – DVC

Weather data changes continuously over time.

DVC is used to manage and version control datasets so that:

- Every change in data is tracked
- The exact dataset used for training is known
- Old and new datasets can be compared when performance drops

This is important because weather prediction fails when new data is very different from training data.



Phase II – MLOPs Life Cycle

Detailed Explanation

Experiment Tracking & Version Control – ZenML

During model development, multiple experiments are performed with different parameters and datasets.

ZenML helps by:

- Tracking model versions
- Storing parameters and evaluation metrics
- Maintaining experiment history

This ensures we always know which model version is the best and deployed.



Phase II – MLOPs Life Cycle

Detailed Explanation

```
Registered new pipeline: weather_pipeline.  
Using user: default  
Using stack: default  
  orchestrator: default  
  artifact_store: default  
  deployer: default  
You can visualize your pipeline runs in the ZenML Dashboard. In order to try it locally, please run zenml login --local.  
Step load_data has started.  
Preparing to run step load_data.  
Step load_data has finished in 12.117s.  
Step preprocess_data has started.  
Preparing to run step preprocess_data.  
Step preprocess_data has finished in 16.915s.  
Step train_model has started.  
Preparing to run step train_model.  
Step train_model has finished in 11.194s.  
Step evaluate_model has started.  
Preparing to run step evaluate_model.  
Step evaluate_model has finished in 1.484s.  
Pipeline run has finished in 48.045s.
```

Fig. ZenML Pipeline



Phase II – MLOPs Life Cycle

Detailed Explanation

Automation & Pipeline Orchestration – ZenML

Manual ML workflows are error-prone.

Using ZenML pipelines, the entire workflow is automated:

- Data loading
- Training
- Evaluation

Whenever new data arrives, the pipeline can be rerun in a structured and reproducible way.



Phase II – MLOPs Life Cycle

Detailed Explanation

Model Deployment & Serving – Docker and FastAPI

Once the best model is selected:

- Docker packages the trained model along with the FastAPI application

- Ensures the selected model version is consistently used during deployment

- FastAPI exposes prediction endpoints for real-time inference

- Docker maintains consistency between training artifacts and the production environment

This avoids deploying outdated or incorrect models.



Phase II – MLOPs Life Cycle

Detailed Explanation

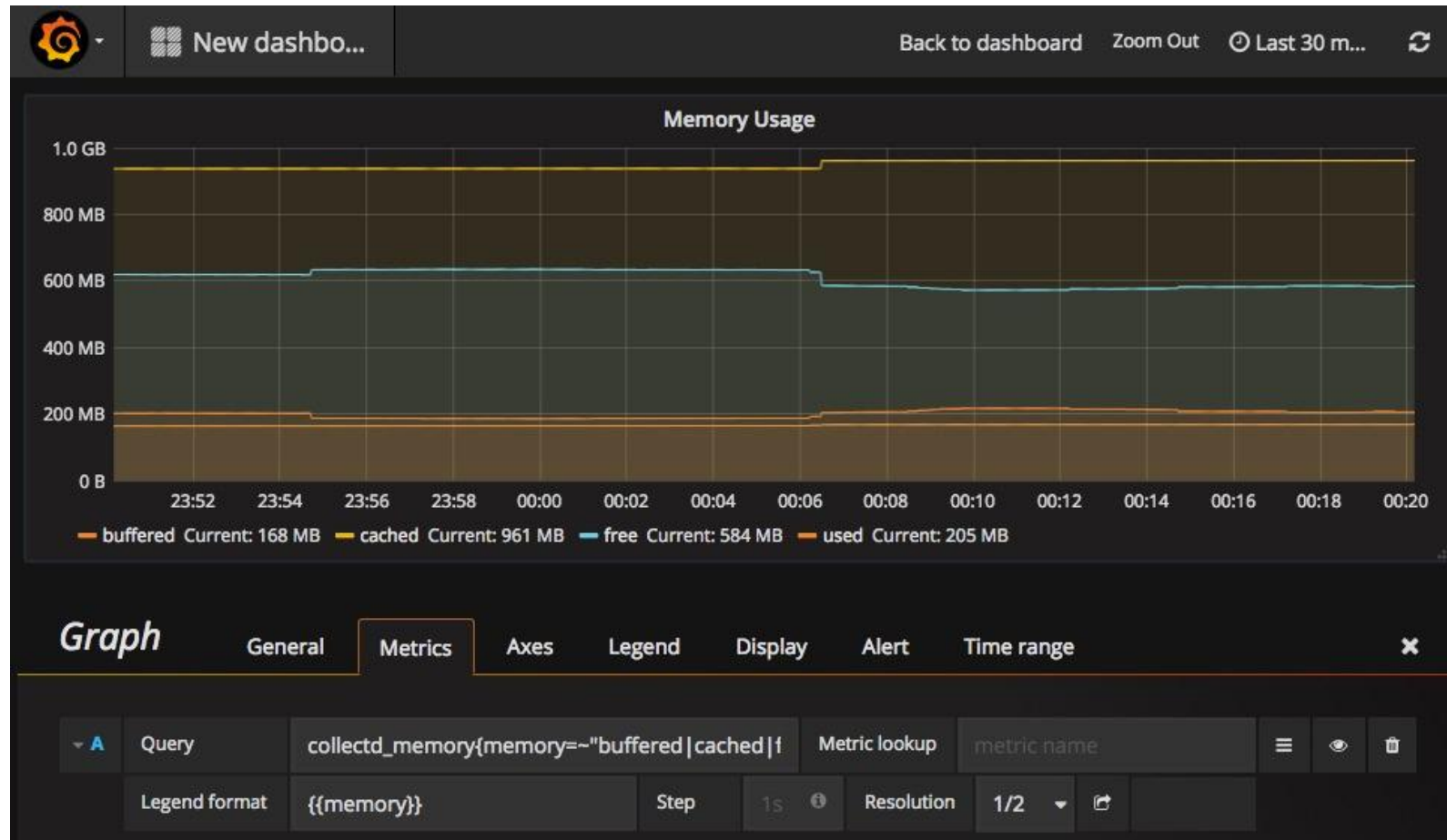


Fig. Prometheus implemtnation



Phase II – MLOPs Life Cycle

Detailed Explanation

Monitoring & Governance – Evidently AI

Weather data changes cause data drift, leading to poor predictions.

Evidently AI is used to:

- Compare incoming data with training data
- Detect feature drift and prediction drift
- Generate drift reports

Instead of automatic retraining, it alerts users, supporting governance and human decision-making.

Phase II – MLOPs Life Cycle

Detailed Explanation

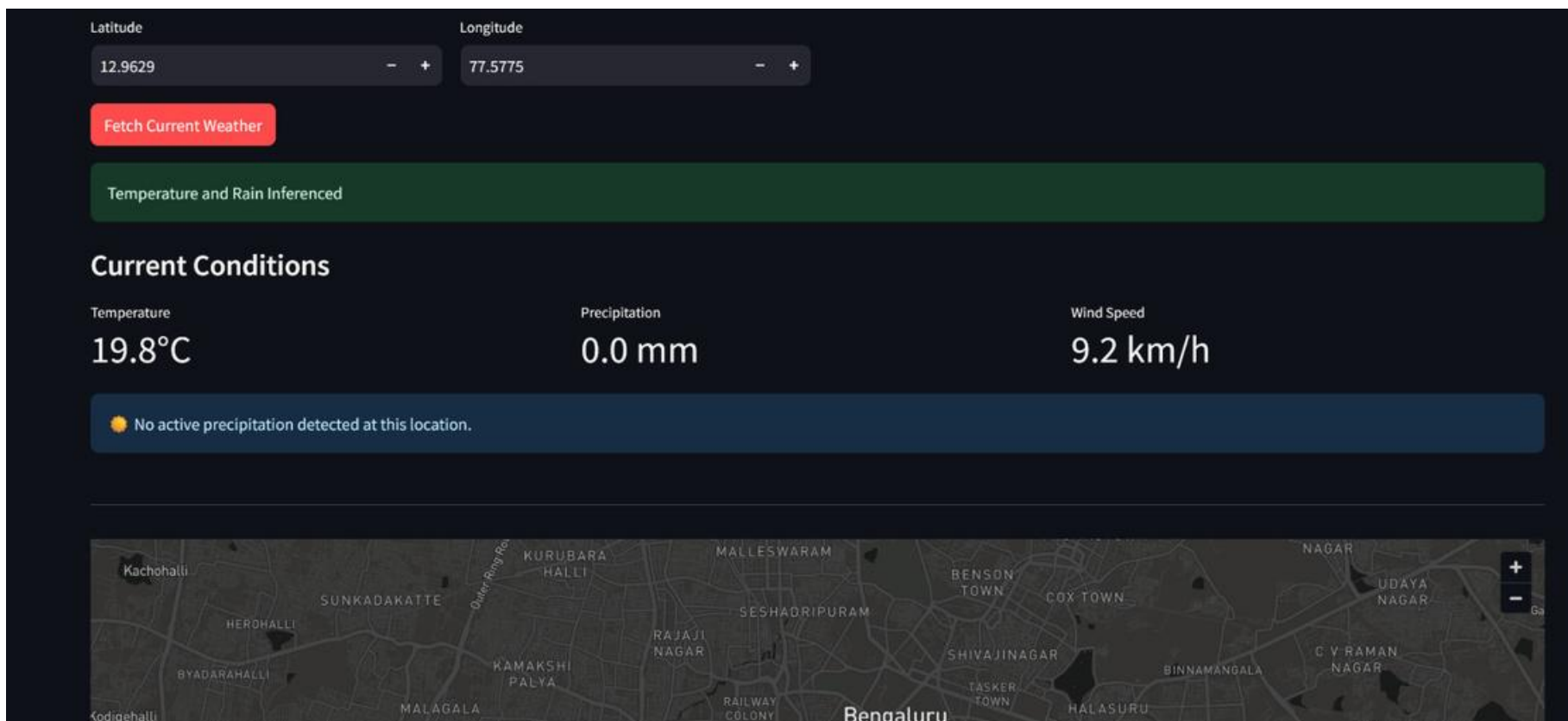


Fig. WebApp interface



Phase II – MLOPs Life Cycle

Conclusion

- Weather prediction models often degrade over time because real-world weather data changes and no longer matches the training data.
- This project shows that machine learning models must be continuously monitored, not treated as one-time deployments.
- An end-to-end MLOps pipeline helps manage data changes, automate workflows, and maintain traceability across the ML lifecycle.
- Data drift detection and user alerts enable informed governance decisions, ensuring long-term reliability of the weather prediction system.